Fabric Inspection Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fabric checking machine

What it is used for

A fabric inspection machine examines textile rolls for defects, shade variation, stains, holes, weaving faults, and dimensional irregularities. Correct lighting, controlled fabric movement, and systematic defect recording help maintain product quality and reduce downstream rejection.

Main parts

- Fabric unwinding roller and roll holder
- Inspection table or illuminated viewing surface
- Fabric take-up roller or winding unit
- Drive motor, speed controls, and tension adjustment
- Lighting system and inspection scale
- Emergency stop, guards, and defect-marking supplies

Before you start

1. Confirm the fabric specification, roll identification, inspection criteria, and acceptable defect limits.
2. Inspect the rollers, guides, inspection surface, lighting, guards, and drive components for damage.
3. Ensure the inspection area is clean and the lighting is suitable for identifying the fabric's specified defects.
4. Check that the fabric roll is securely mounted and compatible with the machine's width and load capacity.
5. Verify that the emergency stop and speed controls function correctly.
6. Prepare the defect log, marking tools, and required PPE; remove jewellery or loose items that could catch in moving fabric.

How to use

1. Switch off the machine before mounting the roll or adjusting roller alignment and fabric guides.
2. Load the fabric roll onto the unwinding holder and secure it using the approved method.
3. Thread the fabric through the guides and inspection area according to the machine diagram.
4. Set fabric tension and inspection speed to suit the material and inspection procedure.
5. Switch on the inspection lighting and confirm that the viewing surface is evenly illuminated.
6. Start the machine at a low speed and check that the fabric tracks centrally without wrinkles or excessive tension.

7. Increase speed gradually to the approved inspection rate while maintaining clear visibility of the fabric.
8. Inspect the full fabric width for holes, stains, slubs, broken yarns, shade differences, and other specified defects.
9. Mark defects using the approved method without damaging the fabric and record their positions or lengths.
10. Stop the machine safely before correcting tracking problems, removing trapped material, or adjusting tension.
11. Continue inspection until the full roll has passed the viewing area and the required length has been checked.
12. Review the defect record and classify the roll according to the applicable quality criteria.
13. Stop the machine, isolate power where required, and remove the inspected roll safely.
14. Clean the inspection surface and record roll status, defect details, and any equipment problems.

⚠ Safety

- Only trained operators should use the fabric inspection machine.
- Keep hands, hair, and loose clothing away from rollers, winding shafts, and pinch points.
- Never reach between moving rollers or attempt to remove trapped fabric while the machine is running.
- Use the emergency stop if the fabric becomes dangerously tangled or an unsafe condition develops.
- Do not exceed the machine's rated roll weight, width, or operating speed.
- Isolate power before maintenance, cleaning near moving components, or correcting mechanical faults.
- Use safe lifting methods or suitable lifting equipment for heavy fabric rolls.

Care & storage

- Clean the inspection table, rollers, and viewing surface after use.
- Inspect rollers, belts, tension controls, and guards for wear or damage.
- Keep inspection lights clean and replace faulty lamps according to the manufacturer's instructions.
- Check speed controls and emergency stops at the required intervals.
- Arrange servicing if fabric tracking becomes unstable, winding tension varies, or inspection lighting fails.

Common mistakes to avoid

- Inspecting fabric under poor or uneven lighting.
- Running the fabric too quickly to identify defects accurately.
- Applying excessive tension and stretching or distorting the fabric.
- Failing to record or mark defects consistently.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.